

KRISHNA Koushik Thokala

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Summary

Infrastructure engineer with 2+ years of production network operations experience in 24/7 Tier-1 carrier environments, including incident response, change execution, root cause analysis, and SLA-driven restoration. Hands-on AWS experience building Terraform-managed EKS and serverless infrastructure, GitOps/CI/CD pipelines, Python automation, and Prometheus/CloudWatch observability. AWS Certified Solutions Architect – Associate; MS in Computer Science.

Technical Skills

Cloud & IaC: AWS (VPC, EC2, EKS, Lambda, RDS, DynamoDB, S3, IAM), Terraform
Containers & Delivery: Kubernetes, Docker, Helm, Argo CD, GitHub Actions, GitOps, CI/CD, ECR
Observability & Reliability: Prometheus, Grafana, CloudWatch, SLIs/SLOs, burn-rate alerting, k6
Systems & Automation: Linux, Python, Bash, Git, SQL
Networking & Security: TCP/IP, DNS, BGP, MPLS, OIDC, RBAC, NetworkPolicies, Secrets Manager

Experience

Tata Communications

July 2022 – July 2024

Senior Engineer, Network Operations Center – Shift Lead

Pune, India

Global telecommunications provider supporting Tier-1 carriers across IP/MPLS and optical transport networks

- Led five-engineer shifts in a 24/7 NOC, prioritizing 40+ daily incidents across 25+ Tier-1 carrier clients while supporting 99.9% availability targets and four-hour restoration SLAs; trained engineers joining the rotation.
- Led P1/P2 incident response in ServiceNow, isolating faults across BGP/MPLS, TCP/IP, DNS, DWDM, and hardware while coordinating vendor TAC, field teams, customer testing, and client communication through restoration.
- Executed 4+ planned network changes per month during scheduled maintenance windows, validating service state before and after each change and initiating rollback when verification failed.
- Authored runbooks, post-incident RCAs, and standardized shift handoffs so incoming engineers inherited complete diagnostic context; earned a Certificate of Excellence for resolving an escalated customer case through closure.

Cloud Infrastructure Projects

EKS Reliability Platform | EKS, Terraform, Helm, Prometheus, k6, Python/FastAPI, RDS

GitHub

- Built and deployed a reliability-operations platform on AWS EKS using Terraform and Helm, with a FastAPI/React application and private RDS PostgreSQL tracking service health, incidents, MTTR, SLOs, and error budgets.
- Instrumented route-template Prometheus metrics to bound label cardinality; implemented availability/latency SLIs and fast/slow multi-window burn-rate alerts for a 99.9% availability objective, with Prometheus rule tests in CI.
- Validated HPA elasticity on EKS with 18,819 k6 requests: the backend scaled 2 → 6 → 2 replicas with zero request failures and zero pod restarts.

GitOps Delivery Platform | EKS, Argo CD, Terraform, Helm, GitHub Actions, OIDC

GitHub

- Built an EKS GitOps delivery platform with Argo CD, Helm, Terraform, and GitHub Actions OIDC; promoted immutable ECR images through Git-based environment changes with namespace-scoped RBAC, ResourceQuotas, and NetworkPolicies across dev, staging, and prod.
- Validated reconciliation and recovery on EKS: Argo CD self-healed manual replica drift, and a Git revert restored a broken-readiness staging release from Degraded to Synced and Healthy.

Serverless Uptime Monitor | Lambda, EventBridge, DynamoDB, API Gateway, CloudWatch

Live | GitHub

- Built a Terraform-managed serverless uptime monitor using Lambda, EventBridge, DynamoDB, API Gateway, CloudWatch, SNS, and CloudFront; redesigned latest-status access to read 10 current-state records instead of scanning 86,400 retained records for a 10-URL/30-day workload.
- Hardened outbound checks against SSRF by rejecting private, local, reserved, metadata, and unsafe redirect targets; a live AWS drill observed one-second failure detection, dashboard DOWN visibility in 26 seconds, no duplicate DOWN alert on the repeated failure check, and recovery notification.

Education

Indiana University Bloomington

August 2024 – May 2026

Master of Science in Computer Science, GPA: 3.5/4.0

Bloomington, IN

SRM Institute of Science and Technology

June 2018 – June 2022

Bachelor of Technology in Computer Science and Engineering

Chennai, India